

Parametric Survival Modeling

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents parametric survival models fitted for the METIMMOX-1 economic evaluation. The models compare different biomarker strategies for overall survival (OS) and progression-free survival (PFS) outcomes. This analysis serves as the foundation for the cost-effectiveness evaluation by quantifying treatment effects across biomarker-defined patient subgroups.

1.1 Model Structure Types

This report compares three model structure types to assess sensitivity to biomarker correlation and model complexity:

- **Model A (Joint):** All three biomarkers and their treatment interactions are included in a single model. This specification tests each biomarker interaction while adjusting for the others, providing mutually adjusted effect estimates. Model A is the primary specification used in the cost-effectiveness analysis.
- **Model B (Focused):** Per-strategy formulas. CRP and TMB/BRAF strategies share a formula with both biomarkers as main effects and both interaction terms (`crp:Rx + tmb_braf:Rx`). TLR uses a separate formula with CRP and TMB/BRAF as main effects but only the TLR interaction (`tlr:Rx`). This creates three models per outcome (B-CRP, B-TLR, B-TMB).
- **Model C (Single-biomarker):** Each biomarker is modeled separately without adjustment for other biomarkers, creating three independent models per outcome (C-CRP, C-TLR, C-TMB). This is the simplest specification but may overestimate effects if biomarkers are correlated.

For each model structure type, we fit two parametric distributions (Gamma and Weibull proportional hazards) for both OS and PFS, resulting in **28 total models** (7 model variants $\times$ 2 distributions $\times$ 2 outcomes).

2 Population Characteristics including Biomarker Prevalences

Population characteristics not available. Run main analysis first.

3 Model Selection and Comparison

3.1 Analysis Specifications

3.1.1 Model Structure Types

We compare seven model variants representing three structural approaches:

Model Type	Structure	Formula
Model A	Joint	Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx
Model B-CRP/TMB	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx
Model B-TLR	Focused	Age + sex + Rx + crp + tmb_braf + tlr:Rx
Model C-CRP	Single-biomarker	Age + sex + Rx + crp:Rx
Model C-TLR	Single-biomarker	Age + sex + Rx + tlr:Rx
Model C-TMB	Single-biomarker	Age + sex + Rx + tmb_braf:Rx

All models include:

- **Age** (continuous, years) - prognostic covariate

- **Sex** (categorical, 0=female, 1=male) - prognostic covariate
- **Treatment** (Rx) - experimental vs control
- **Biomarker interactions** - as shown in formulas above

3.1.2 Distributions Tested

For Model A (Joint), we test nine parametric distributions to select the best-fitting family:

1. Exponential
2. Weibull
3. Weibull PH
4. Gompertz
5. Log-logistic
6. Log-normal
7. Generalized Gamma
8. Gamma
9. Generalized F Distribution

The best-fitting distribution (Gamma) and reference distribution (Weibull PH) are then applied to all seven model structure types.

3.1.3 Data Considerations

- **Complete case analysis:** Only patients with complete covariate data included
- **Time horizon:** Models fitted to observed follow-up period with extrapolation capability
- **Biomarker prevalences:** Taken from the METIMMOX-1 trial and may or may not reflect real world settings

3.1.4 Model Selection Criterion

AIC (Akaike Information Criterion) is used for model selection because:

- BIC penalizes model complexity more heavily (uses $\log(n)$ vs 2 for AIC)
- For prediction purposes, we want the model that best balances fit and complexity
- AIC is better for predictive modeling while BIC is better for identifying the “true” model
- Validation via visual comparison with Kaplan-Meier curves and clinical plausibility assessment

3.2 Distribution Selection (Model A)

We first select the best-fitting parametric distribution using Model A (Joint), the most comprehensive specification. This distribution is then applied to all seven model structure types for fair comparison.

Table 1: Overall Survival (OS) distribution comparison for Model A using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	622.56	646.47	0.00	1.0	***
Full Model	weibull	623.44	647.36	0.88	2.5	
Full Model	weibullph	623.44	647.36	0.88	2.5	
Full Model	llogis	623.68	647.60	1.12	4.0	
Full Model	lognormal	624.09	648.01	1.53	5.0	
Full Model	gengamma	624.55	650.65	1.99	6.0	
Full Model	genf	626.56	654.82	4.00	7.0	
Full Model	gompertz	633.09	657.01	10.53	8.0	
Full Model	exponential	654.14	675.88	31.58	9.0	

Table 2: Progression-Free Survival (PFS) distribution comparison for Model A using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	453.43	477.35	0.00	1.0	***
Full Model	lognormal	453.46	477.38	0.03	2.0	
Full Model	weibull	454.19	478.11	0.76	3.5	
Full Model	weibullph	454.19	478.11	0.76	3.5	
Full Model	llogis	455.16	479.08	1.73	5.0	
Full Model	gengamma	455.28	481.37	1.85	6.0	
Full Model	genf	457.28	485.55	3.85	7.0	
Full Model	gompertz	467.57	491.49	14.14	8.0	
Full Model	exponential	483.52	505.26	30.09	9.0	

Based on AIC, **Gamma** is selected as the best-fitting distribution for both OS and PFS. **Weibull PH** is retained as a reference distribution for robustness assessment.

3.3 Model Structure Comparison

Having selected Gamma as the best-fitting distribution, we now compare all seven model structure types using this distribution. This comparison assesses the impact of biomarker specification on model fit.

Table 3: Model structure comparison using Gamma distribution

	Model Type	OS AIC	OS Delta AIC	OS BIC	PFS AIC	PFS Delta AIC	PFS BIC
A	Model A	622.56	3.90	646.47	453.43	2.36	477.35
B_crp	Model B-CRP	620.08	1.42	639.64	451.07	0.00	470.64
B_tlr	Model B-TLR	618.66	0.00	638.23	452.23	1.16	471.80
B_tmb	Model B-TMB	620.05	1.39	639.62	451.74	0.67	471.30
C_crp	Model C-CRP	627.26	8.60	642.48	481.70	30.63	496.92
C_tlr	Model C-TLR	620.54	1.88	635.76	463.40	12.33	478.62
C_tmb	Model C-TMB	635.54	16.88	650.76	487.25	36.18	502.47

Interpretation:

- **Model A (Joint)** has the lowest AIC (most parameters, best fit) but highest BIC (penalized for complexity)
- **Model C variants** (Single-biomarker) have the highest AIC (fewest parameters, worse fit) but lowest BIC

- **Model B variants** (Focused) represent intermediate complexity
- All Delta AIC values are relatively small, indicating reasonable fit across all model structures
- **Model A is used for the primary cost-effectiveness analysis** as it provides the most comprehensive assessment of biomarker interactions while controlling for confounding

4 Model Coefficients

The following tables present model coefficients for each model type, organized by outcome and distribution. Coefficients are presented as estimates with 95% confidence intervals. For treatment-biomarker interaction hazard ratios, see the Clinical Effectiveness report.

4.1 Model A (Joint) Coefficients

Model A includes all three biomarker interactions: `Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx`

4.1.1 Overall Survival

Gamma Distribution:

Table 4: Model A - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2275	0.1820	0.8707	1.5843
rate	-2.8975	0.6157	-4.1041	-1.6908
Age	0.0015	0.0082	-0.0145	0.0175
sex1	0.1454	0.1482	-0.1450	0.4358
RxExperimental arm	-0.2309	0.2852	-0.7898	0.3281
RxControl arm:crp1	-0.3570	0.3162	-0.9767	0.2628
RxExperimental arm:crp1	-0.3395	0.2265	-0.7834	0.1044
RxControl arm:tlr1	-0.8039	0.2509	-1.2956	-0.3121
RxExperimental arm:tlr1	-0.4109	0.2129	-0.8282	0.0064
RxControl arm:tmb_braf1	-0.0866	0.2210	-0.5197	0.3465
RxExperimental arm:tmb_braf1	-0.1859	0.2072	-0.5920	0.2202

Weibull PH Distribution:

Table 5: Model A - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7281	0.1090	0.5145	0.9417
scale	-9.3901	1.5795	-12.4857	-6.2944
Age	0.0105	0.0168	-0.0224	0.0434
sex1	0.3426	0.2988	-0.2430	0.9282
RxExperimental arm	-0.4313	0.5275	-1.4652	0.6026
RxControl arm:crp1	-0.8624	0.6443	-2.1252	0.4005
RxExperimental arm:crp1	-0.6574	0.4639	-1.5667	0.2518
RxControl arm:tlr1	-1.6010	0.4921	-2.5656	-0.6365
RxExperimental arm:tlr1	-0.7773	0.4178	-1.5961	0.0415
RxControl arm:tmb_braf1	-0.1394	0.4303	-0.9828	0.7040
RxExperimental arm:tmb_braf1	-0.2591	0.4186	-1.0797	0.5614

4.1.2 Progression-Free Survival

Gamma Distribution:

Table 6: Model A - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2426	0.1901	0.8701	1.6152
rate	-1.8647	0.6321	-3.1036	-0.6258
Age	-0.0009	0.0084	-0.0173	0.0155
sex1	0.1321	0.1653	-0.1919	0.4562
RxExperimental arm	0.4218	0.3073	-0.1804	1.0240
RxControl arm:crp1	0.0654	0.4253	-0.7681	0.8990
RxExperimental arm:crp1	-0.5754	0.2412	-1.0481	-0.1026
RxControl arm:tlr1	-0.9764	0.2718	-1.5091	-0.4436
RxExperimental arm:tlr1	-1.1914	0.2133	-1.6095	-0.7732
RxControl arm:tmb_braf1	-0.1055	0.2537	-0.6027	0.3917
RxExperimental arm:tmb_braf1	-0.4891	0.2325	-0.9447	-0.0334

Weibull PH Distribution:

Table 7: Model A - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7437	0.1139	0.5206	0.9669
scale	-6.7666	1.3964	-9.5034	-4.0298
Age	-0.0032	0.0169	-0.0362	0.0299
sex1	0.2264	0.3463	-0.4524	0.9052
RxExperimental arm	0.9743	0.5936	-0.1892	2.1378
RxControl arm:crp1	0.4137	0.8152	-1.1841	2.0114
RxExperimental arm:crp1	-1.0969	0.5515	-2.1779	-0.0160
RxControl arm:tlr1	-2.0111	0.5566	-3.1020	-0.9203
RxExperimental arm:tlr1	-2.3671	0.4623	-3.2732	-1.4610
RxControl arm:tmb_braf1	-0.2783	0.4940	-1.2465	0.6900
RxExperimental arm:tmb_braf1	-1.2349	0.5657	-2.3438	-0.1261

4.2 Model B-CRP (Focused) Coefficients

Model B-CRP includes CRP and TMB/BRAF with both interaction terms: Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx

4.2.1 Overall Survival

Gamma Distribution:

Table 8: Model B-CRP - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2014	0.1816	0.8454	1.5574
rate	-3.1792	0.5893	-4.3342	-2.0242
Age	0.0030	0.0082	-0.0130	0.0190
sex1	0.1671	0.1489	-0.1247	0.4590
RxExperimental arm	-0.0460	0.1942	-0.4267	0.3348
crp1	-0.4286	0.3141	-1.0443	0.1871
tlr1	-0.5728	0.1679	-0.9020	-0.2437
tmb_braf1	-0.1393	0.1537	-0.4406	0.1620
RxExperimental arm:crp1	0.1362	0.3801	-0.6088	0.8812

Weibull PH Distribution:

Table 9: Model B-CRP - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7108	0.1089	0.4973	0.9243
scale	-9.7057	1.5689	-12.7806	-6.6307
Age	0.0125	0.0165	-0.0198	0.0448
sex1	0.3834	0.2950	-0.1948	0.9616
RxExperimental arm	-0.0254	0.3810	-0.7722	0.7214
crp1	-0.9874	0.6312	-2.2245	0.2497
tlr1	-1.1070	0.3393	-1.7721	-0.4420
tmb_braf1	-0.1965	0.3008	-0.7862	0.3931
RxExperimental arm:crp1	0.4682	0.7503	-1.0024	1.9389

4.2.2 Progression-Free Survival

Gamma Distribution:

Table 10: Model B-CRP - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2150	0.1897	0.8433	1.5868
rate	-1.6798	0.5966	-2.8492	-0.5105
Age	-0.0011	0.0086	-0.0180	0.0157
sex1	0.1155	0.1645	-0.2070	0.4380
RxExperimental arm	0.1688	0.1969	-0.2171	0.5548
crp1	0.0138	0.4205	-0.8103	0.8378
tlr1	-1.1138	0.1681	-1.4432	-0.7844
tmb_braf1	-0.3234	0.1742	-0.6647	0.0180
RxExperimental arm:crp1	-0.7043	0.4852	-1.6552	0.2466

Weibull PH Distribution:

Table 11: Model B-CRP - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7183	0.1130	0.4967	0.9398
scale	-6.3517	1.3605	-9.0183	-3.6852
Age	-0.0009	0.0175	-0.0352	0.0335
sex1	0.1644	0.3334	-0.4890	0.8179
RxExperimental arm	0.5028	0.3809	-0.2438	1.2493
crp1	0.3007	0.7921	-1.2517	1.8531
tlr1	-2.2001	0.3841	-2.9531	-1.4472
tmb_braf1	-0.7223	0.3729	-1.4532	0.0086
RxExperimental arm:crp1	-1.7545	0.9683	-3.6523	0.1432

4.3 Model B-TMB (Focused) Coefficients

Model B-TMB includes CRP and TMB/BRAF with both interaction terms: Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx

4.3.1 Overall Survival

Gamma Distribution:

Table 12: Model B-TMB - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2045	0.1816	0.8487	1.5604
rate	-3.2426	0.5630	-4.3459	-2.1392
Age	0.0032	0.0080	-0.0125	0.0189
sex1	0.1698	0.1484	-0.1211	0.4607
RxExperimental arm	0.0325	0.2017	-0.3629	0.4278
crp1	-0.3189	0.1819	-0.6755	0.0376
tlr1	-0.5689	0.1675	-0.8971	-0.2406
tmb_braf1	-0.0623	0.2211	-0.4957	0.3711
RxExperimental arm:tmb_braf1	-0.1167	0.3039	-0.7123	0.4790

Weibull PH Distribution:

Table 13: Model B-TMB - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7046	0.1085	0.4920	0.9172
scale	-9.8021	1.5466	-12.8334	-6.7708
Age	0.0131	0.0165	-0.0192	0.0454
sex1	0.3901	0.2940	-0.1861	0.9664
RxExperimental arm	0.1300	0.3957	-0.6456	0.9056
crp1	-0.6548	0.3602	-1.3609	0.0512
tlr1	-1.0772	0.3387	-1.7411	-0.4134
tmb_braf1	-0.0925	0.4277	-0.9307	0.7458
RxExperimental arm:tmb_braf1	-0.1026	0.5895	-1.2580	1.0528

4.3.2 Progression-Free Survival

Gamma Distribution:

Table 14: Model B-TMB - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.1973	0.1898	0.8253	1.5694
rate	-1.5557	0.5761	-2.6849	-0.4265
Age	-0.0034	0.0083	-0.0198	0.0129
sex1	0.0975	0.1666	-0.2290	0.4240
RxExperimental arm	0.1810	0.2179	-0.2460	0.6080
crp1	-0.4579	0.2222	-0.8934	-0.0223
tlr1	-1.1052	0.1680	-1.4346	-0.7759
tmb_braf1	-0.1908	0.2500	-0.6808	0.2993
RxExperimental arm:tmb_braf1	-0.3880	0.3472	-1.0686	0.2925

Weibull PH Distribution:

Table 15: Model B-TMB - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7114	0.1136	0.4888	0.9341
scale	-6.0352	1.2844	-8.5526	-3.5178
Age	-0.0056	0.0165	-0.0380	0.0267
sex1	0.1264	0.3369	-0.5340	0.7868
RxExperimental arm	0.5340	0.4170	-0.2834	1.3514
crp1	-0.7508	0.4864	-1.7040	0.2025
tlr1	-2.1477	0.3740	-2.8806	-1.4147
tmb_braf1	-0.4119	0.4765	-1.3458	0.5220
RxExperimental arm:tmb_braf1	-1.0639	0.7247	-2.4844	0.3565

4.4 Model B-TLR (Focused) Coefficients

Model B-TLR includes CRP and TMB/BRAF as main effects with TLR interaction only: Age + sex + Rx + crp + tmb_braf + tlr:Rx

4.4.1 Overall Survival

Gamma Distribution:

Table 16: Model B-TLR - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.2262	0.1819	0.8696	1.5828
rate	-2.8959	0.6110	-4.0935	-1.6983
Age	0.0018	0.0081	-0.0140	0.0176
sex1	0.1448	0.1482	-0.1457	0.4353
RxExperimental arm	-0.2684	0.2562	-0.7706	0.2337
crp1	-0.3607	0.1769	-0.7074	-0.0141
tlr1	-0.8085	0.2466	-1.2919	-0.3251
tmb_braf1	-0.1390	0.1478	-0.4286	0.1507
RxExperimental arm:tlr1	0.4020	0.3156	-0.2166	1.0206

Weibull PH Distribution:

Table 17: Model B-TLR - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.7277	0.1089	0.5142	0.9412
scale	-9.4206	1.5784	-12.5142	-6.3269
Age	0.0114	0.0165	-0.0209	0.0438
sex1	0.3397	0.2979	-0.2441	0.9235
RxExperimental arm	-0.4571	0.4820	-1.4018	0.4875
crp1	-0.7472	0.3570	-1.4468	-0.0476
tlr1	-1.6265	0.4852	-2.5774	-0.6755
tmb_braf1	-0.1848	0.2926	-0.7583	0.3887
RxExperimental arm:tlr1	0.8748	0.6006	-0.3022	2.0519

4.4.2 Progression-Free Survival

Gamma Distribution:

Table 18: Model B-TLR - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.1936	0.1901	0.8211	1.5661
rate	-1.7112	0.6342	-2.9542	-0.4682
Age	-0.0020	0.0086	-0.0188	0.0148
sex1	0.1282	0.1695	-0.2040	0.4604
RxExperimental arm	0.2367	0.2849	-0.3217	0.7951
crp1	-0.5269	0.2065	-0.9316	-0.1223
tlr1	-0.9278	0.2731	-1.4630	-0.3926
tmb_braf1	-0.3866	0.1693	-0.7184	-0.0548
RxExperimental arm:tlr1	-0.2949	0.3489	-0.9789	0.3890

Weibull PH Distribution:

Table 19: Model B-TLR - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.6854	0.1112	0.4674	0.9033
scale	-6.0582	1.3409	-8.6863	-3.4300
Age	-0.0037	0.0172	-0.0374	0.0299
sex1	0.1948	0.3469	-0.4851	0.8746
RxExperimental arm	0.5766	0.5430	-0.4876	1.6407
crp1	-1.0474	0.4462	-1.9219	-0.1728
tlr1	-1.7860	0.5378	-2.8402	-0.7319
tmb_braf1	-0.8691	0.3665	-1.5874	-0.1508
RxExperimental arm:tlr1	-0.5213	0.6460	-1.7874	0.7449

4.5 Model C-CRP (Single-biomarker) Coefficients

Model C-CRP includes only CRP with its interaction: `Age + sex + Rx + crp:Rx`

4.5.1 Overall Survival

Gamma Distribution:

Table 20: Model C-CRP - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.0625	0.1806	0.7085	1.4164
rate	-3.6281	0.6118	-4.8272	-2.4289
Age	-0.0007	0.0086	-0.0175	0.0162
sex1	0.2152	0.1567	-0.0920	0.5224
RxExperimental arm	0.2387	0.1863	-0.1265	0.6040
RxControl arm:crp1	-0.5503	0.3328	-1.2026	0.1020
RxExperimental arm:crp1	-0.5779	0.2094	-0.9883	-0.1676

Weibull PH Distribution:

Table 21: Model C-CRP - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.6435	0.1101	0.4277	0.8592
scale	-9.4588	1.4917	-12.3825	-6.5350
Age	0.0022	0.0159	-0.0290	0.0334
sex1	0.4660	0.2812	-0.0852	1.0172
RxExperimental arm	0.5586	0.3250	-0.0784	1.1956
RxControl arm:crp1	-1.0948	0.6239	-2.3177	0.1281
RxExperimental arm:crp1	-1.0473	0.3709	-1.7742	-0.3203

4.5.2 Progression-Free Survival

Gamma Distribution:

Table 22: Model C-CRP - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	0.6644	0.1811	0.3095	1.0193
rate	-2.5955	0.7279	-4.0222	-1.1688
Age	-0.0120	0.0109	-0.0334	0.0095
sex1	0.1799	0.2143	-0.2401	0.5999
RxExperimental arm	0.5097	0.2420	0.0354	0.9839
RxControl arm:crp1	-0.1811	0.5265	-1.2131	0.8508
RxExperimental arm:crp1	-1.1866	0.2692	-1.7141	-0.6590

Weibull PH Distribution:

Table 23: Model C-CRP - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.3692	0.1110	0.1516	0.5868
scale	-4.8053	1.1412	-7.0421	-2.5685
Age	-0.0180	0.0158	-0.0490	0.0129
sex1	0.2304	0.3118	-0.3807	0.8416
RxExperimental arm	0.7218	0.3487	0.0384	1.4051
RxControl arm:crp1	-0.1374	0.7518	-1.6109	1.3362
RxExperimental arm:crp1	-1.6882	0.4193	-2.5100	-0.8665

4.6 Model C-TLR (Single-biomarker) Coefficients

Model C-TLR includes only TLR with its interaction: `Age + sex + Rx + tlr:Rx`

4.6.1 Overall Survival

Gamma Distribution:

Table 24: Model C-TLR - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	1.1657	0.1816	0.8097	1.5217
rate	-3.0555	0.6248	-4.2801	-1.8309
Age	0.0016	0.0082	-0.0145	0.0176
sex1	0.1965	0.1504	-0.0983	0.4912
RxExperimental arm	-0.3288	0.2601	-0.8386	0.1809
RxControl arm:tlr1	-0.8676	0.2504	-1.3583	-0.3769
RxExperimental arm:tlr1	-0.5771	0.1995	-0.9682	-0.1860

Weibull PH Distribution:

Table 25: Model C-TLR - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.6909	0.1086	0.4781	0.9036
scale	-9.1326	1.5369	-12.1448	-6.1204
Age	0.0095	0.0161	-0.0220	0.0410
sex1	0.4158	0.2882	-0.1490	0.9806
RxExperimental arm	-0.5646	0.4744	-1.4943	0.3652
RxControl arm:tlr1	-1.7244	0.4823	-2.6697	-0.7790
RxExperimental arm:tlr1	-1.1094	0.3738	-1.8420	-0.3769

4.6.2 Progression-Free Survival

Gamma Distribution:

Table 26: Model C-TLR - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	0.9603	0.1863	0.5952	1.3255
rate	-2.2803	0.7102	-3.6723	-0.8884
Age	-0.0018	0.0096	-0.0207	0.0170
sex1	0.2786	0.1868	-0.0876	0.6448
RxExperimental arm	0.1357	0.3149	-0.4815	0.7530
RxControl arm:tlr1	-0.9346	0.3056	-1.5336	-0.3356
RxExperimental arm:tlr1	-1.4988	0.2329	-1.9553	-1.0422

Weibull PH Distribution:

Table 27: Model C-TLR - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.4913	0.1057	0.2841	0.6985
scale	-5.6371	1.3324	-8.2486	-3.0256
Age	-0.0021	0.0165	-0.0344	0.0302
sex1	0.5445	0.3290	-0.1004	1.1894
RxExperimental arm	0.2210	0.5234	-0.8048	1.2467
RxControl arm:tlr1	-1.4805	0.5154	-2.4906	-0.4704
RxExperimental arm:tlr1	-2.3826	0.4412	-3.2473	-1.5179

4.7 Model C-TMB (Single-biomarker) Coefficients

Model C-TMB includes only TMB_BRAF with its interaction: Age + sex + Rx + tmb_braf:Rx

4.7.1 Overall Survival

Gamma Distribution:

Table 28: Model C-TMB - OS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	0.9591	0.1802	0.6060	1.3123
rate	-3.7193	0.6087	-4.9124	-2.5263
Age	-0.0029	0.0087	-0.0200	0.0141
sex1	0.2265	0.1676	-0.1020	0.5550
RxExperimental arm	0.2211	0.2136	-0.1976	0.6398
RxControl arm:tmb_braf1	0.0419	0.2491	-0.4464	0.5301
RxExperimental arm:tmb_braf1	-0.3408	0.2223	-0.7764	0.0949

Weibull PH Distribution:

Table 29: Model C-TMB - OS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.5675	0.1088	0.3543	0.7806
scale	-8.6165	1.3710	-11.3036	-5.9293
Age	-0.0033	0.0152	-0.0331	0.0265
sex1	0.4015	0.2890	-0.1650	0.9680
RxExperimental arm	0.4958	0.3590	-0.2078	1.1994
RxControl arm:tmb_braf1	0.0711	0.4277	-0.7671	0.9094
RxExperimental arm:tmb_braf1	-0.6050	0.3748	-1.3396	0.1296

4.7.2 Progression-Free Survival

Gamma Distribution:

Table 30: Model C-TMB - PFS - Gamma

Parameter	Estimate	SE	L95%	U95%
shape	0.5538	0.1777	0.2055	0.9022
rate	-2.4378	0.7846	-3.9756	-0.9000
Age	-0.0153	0.0114	-0.0377	0.0071
sex1	0.1446	0.2297	-0.3056	0.5948
RxExperimental arm	0.3144	0.2878	-0.2496	0.8784
RxControl arm:tmb_braf1	-0.1544	0.3495	-0.8395	0.5307
RxExperimental arm:tmb_braf1	-1.0163	0.2917	-1.5881	-0.4446

Weibull PH Distribution:

Table 31: Model C-TMB - PFS - Weibull PH

Parameter	Estimate	SE	L95%	U95%
shape	0.3686	0.1142	0.1448	0.5924
scale	-4.5085	1.1871	-6.8353	-2.1818
Age	-0.0200	0.0157	-0.0509	0.0108
sex1	0.1522	0.3163	-0.4678	0.7722
RxExperimental arm	0.4358	0.3846	-0.3180	1.1896
RxControl arm:tmb_braf1	-0.2957	0.4808	-1.2381	0.6467
RxExperimental arm:tmb_braf1	-1.5535	0.4344	-2.4049	-0.7021

Note: Parameters include distribution shape/scale parameters (e.g., **shape**, **rate** for Gamma; **shape**, **scale** for Weibull) as well as covariate effects. Coefficients are on the log scale; exponentiate to obtain hazard ratios for covariate effects.

5 Population Survival Curves

The following figures compare empirical Kaplan-Meier estimates with predicted survival curves from the selected parametric models. “Population” refers to different patient groups: biomarker-positive patients, biomarker-negative patients, or the entire population. For the control strategy, population curves represent all patients. For biomarker-guided strategies, population-level predictions are weighted by the prevalence of biomarker-positive and biomarker-negative patients in the sample.

Note: For simplicity, population-level validation plots show Model A (Joint) with Gamma distribution only. All model structure types show similar population-level fit; differences emerge primarily at the biomarker subgroup level (presented in subsequent biomarker-specific sections).

5.1 Model Validation: Comparison with Kaplan-Meier Estimates

These figures show how well each tested parametric distribution fits the observed data for the entire study population.

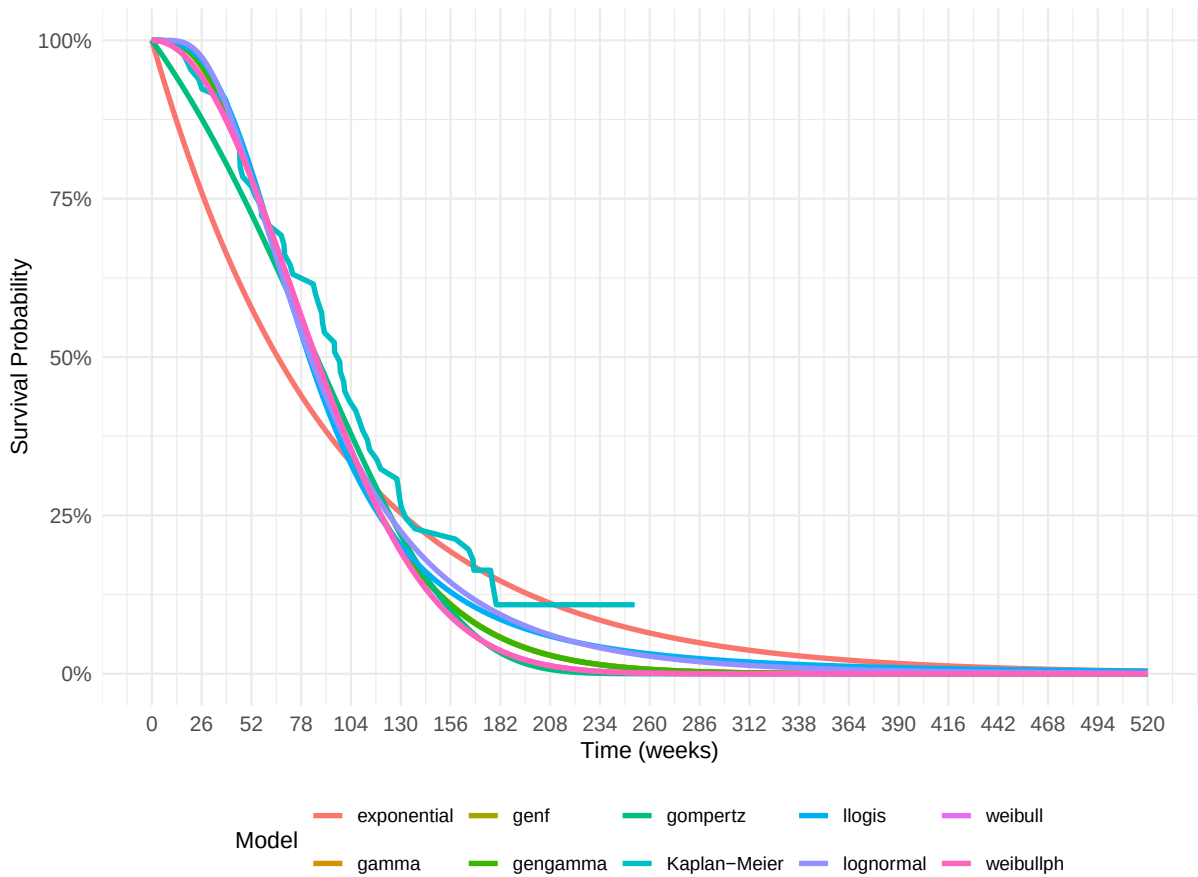


Figure 1: Overall survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

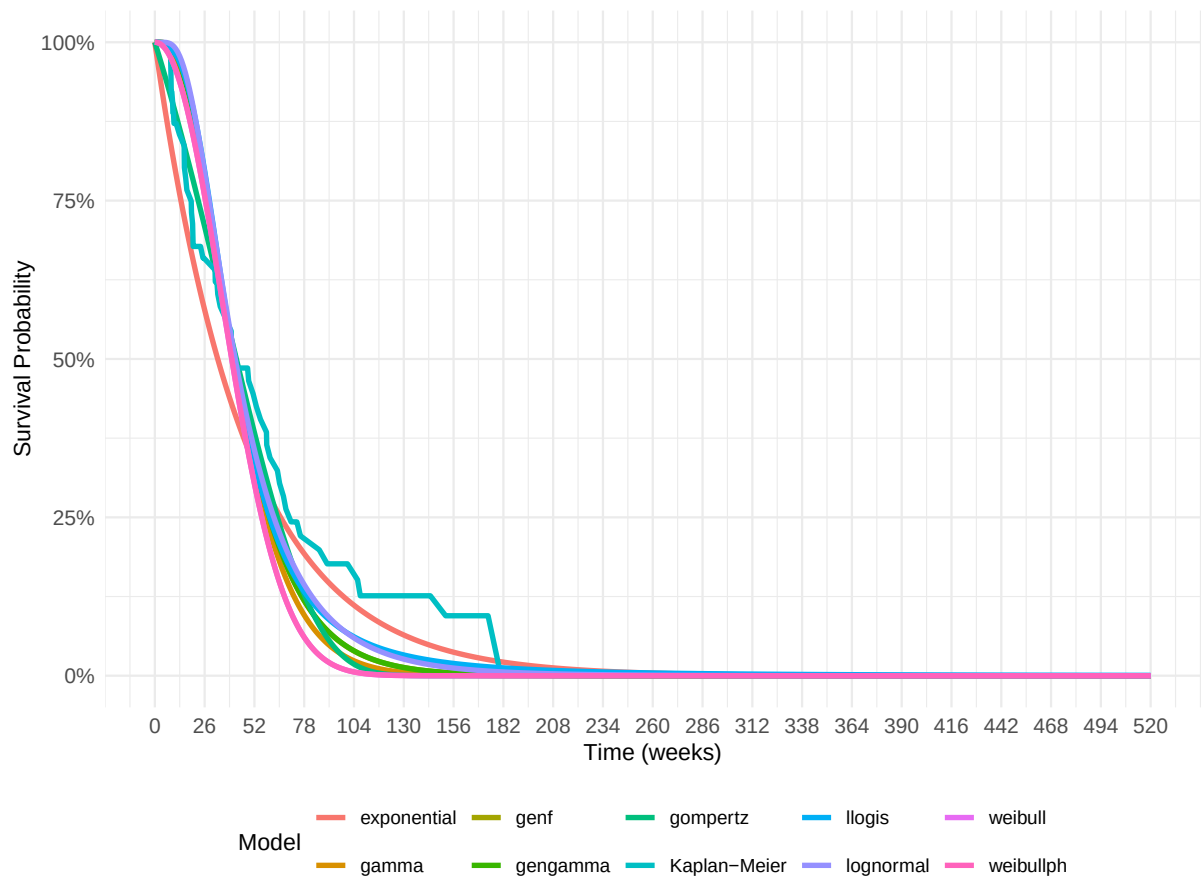


Figure 2: Progression-free survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

6 Biomarker-Specific Results

The following figures compare survival outcomes between biomarker-positive and biomarker-negative patient subgroups for each biomarker strategy. These comparisons inform the potential value of biomarker-guided treatment selection. Kaplan-Meier estimates are shown for biomarker-positive patients receiving experimental treatment and biomarker-negative patients receiving control treatment, alongside parametric model predictions.

6.1 CRP Biomarker Strategy

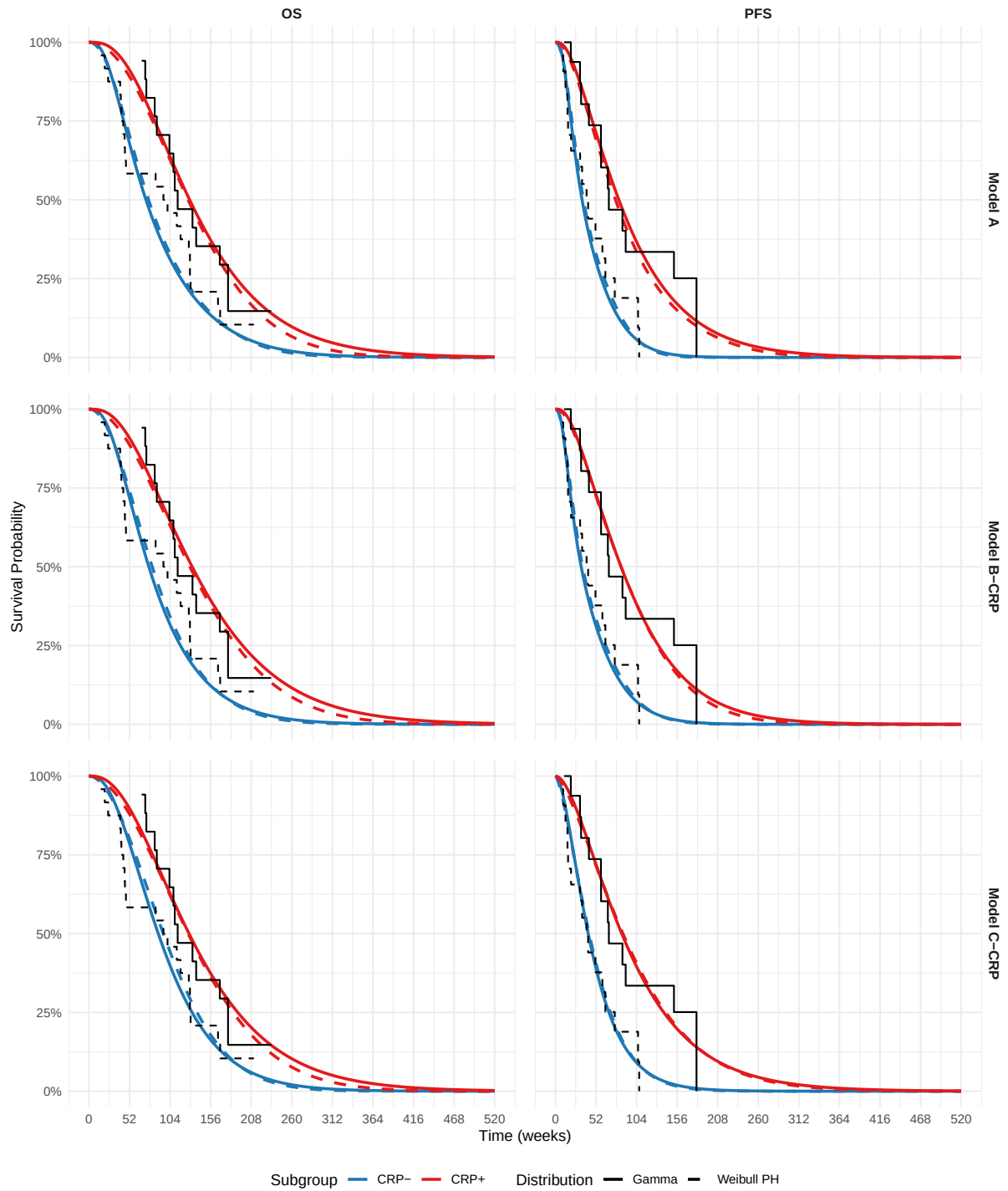


Figure 3: CRP biomarker strategy: Model structure comparison across OS and PFS. Rows show Model A (Joint), Model B-CRP (Focused), and Model C-CRP (Single-biomarker). Colored lines: parametric predictions (red=CRP+, blue=CRP-; solid=Gamma, dashed=Weibull PH). Black step lines: Kaplan-Meier estimates (solid=biomarker+, dashed=biomarker-).

Interpretation: Compare CRP+ (red) and CRP- (blue) survival across model types. Model A shows mutually adjusted effects, Model B-CRP shows effects adjusted for other biomarker main effects, and Model C-CRP shows unadjusted single-biomarker effects. Substantial divergence

between Model C and Models A/B would suggest confounding from other biomarkers.

6.2 TLR Biomarker Strategy

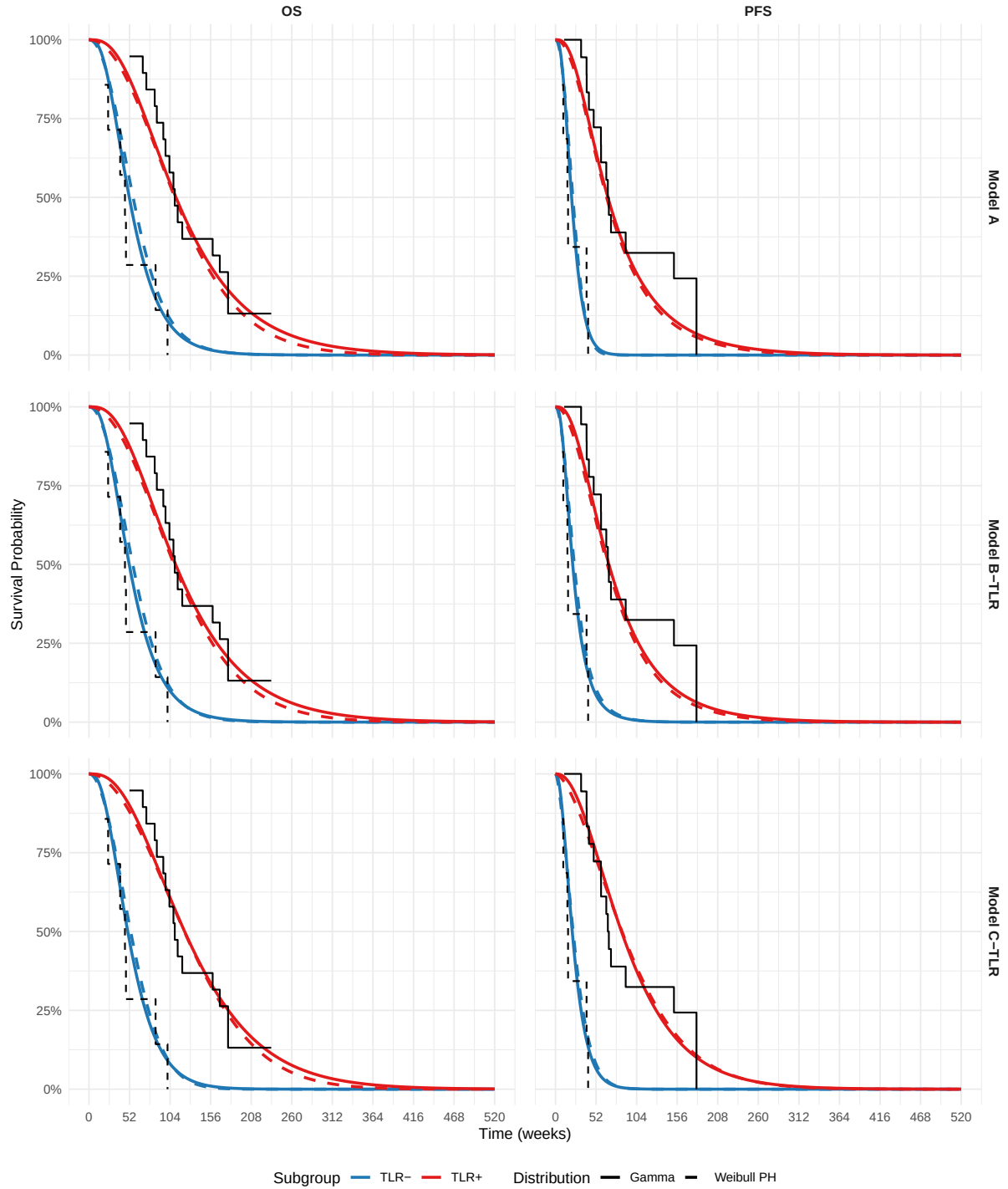


Figure 4: TLR biomarker strategy: Model structure comparison across OS and PFS. Rows show Model A (Joint), Model B-TLR (Focused), and Model C-TLR (Single-biomarker). Colored lines: parametric predictions (red=TLR+, blue=TLR-; solid=Gamma, dashed=Weibull PH). Black step lines: Kaplan-Meier estimates (solid=biomarker+, dashed=biomarker-).

Interpretation: Compare TLR+ (red) and TLR- (blue) survival across model types. TLR (target lesion reduction) is an early response biomarker. Model stability across A, B, and C structures indicates robust predictive value.

6.3 TMB_BRAF Biomarker Strategy

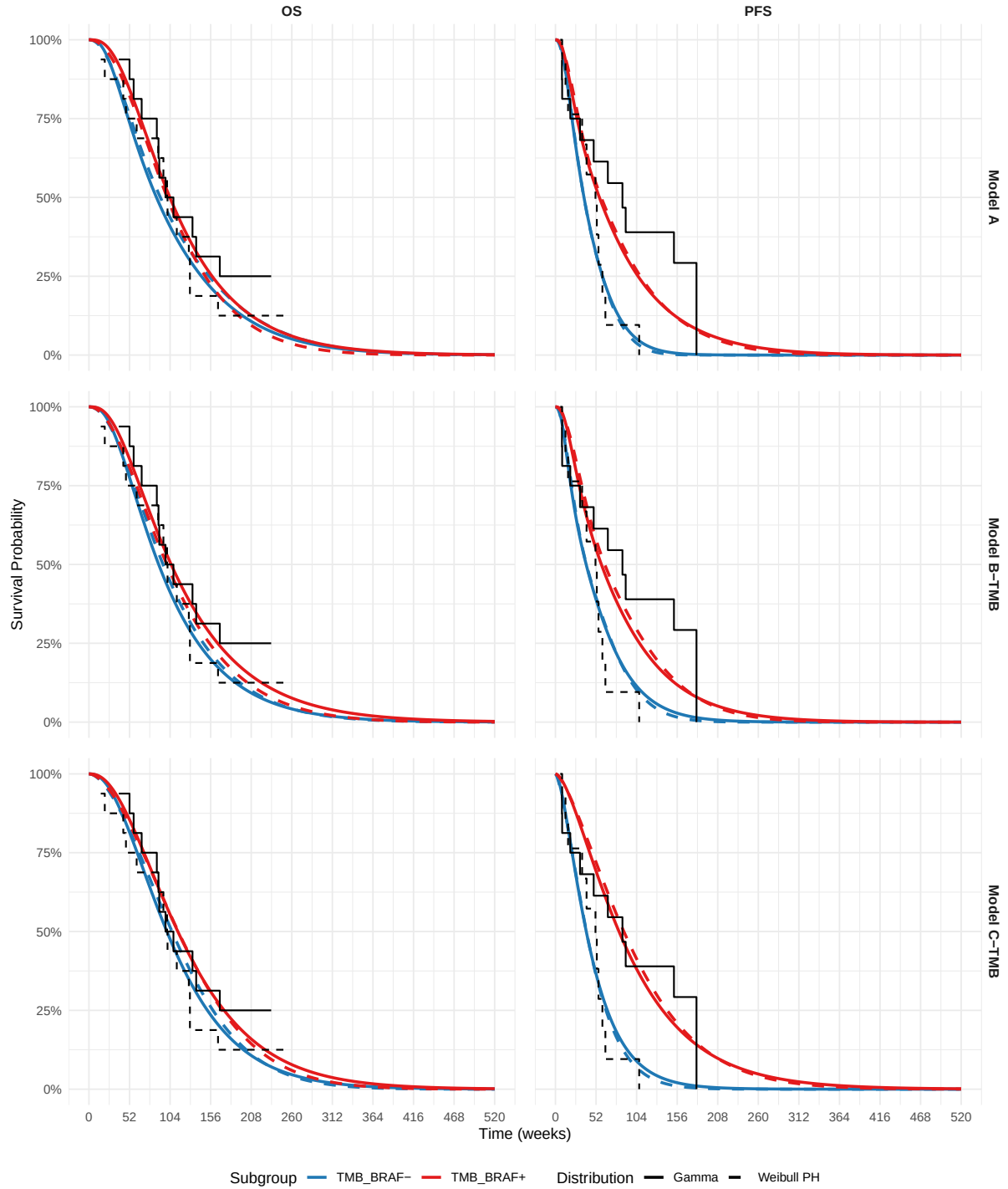


Figure 5: TMB_BRAF biomarker strategy: Model structure comparison across OS and PFS. Rows show Model A (Joint), Model B-TMB (Focused), and Model C-TMB (Single-biomarker). Colored lines: parametric predictions (red=TMB_BRAF+, blue=TMB_BRAF-; solid=Gamma, dashed=Weibull PH). Black step lines: Kaplan-Meier estimates (solid=biomarker+, dashed=biomarker-).

Interpretation: Compare TMB_BRAF+ (red) and TMB_BRAF- (blue) survival across model types. TMB_BRAF is a genomic biomarker combining high tumor mutational burden and

BRAF mutation status. Divergence between model structures may indicate correlation with other biomarkers.

7 Strategy Comparison

The following figures compare biomarker-guided treatment strategies against control, showing how population-weighted survival predictions vary across model structure types. Each biomarker strategy is shown separately to assess sensitivity to model specification.

7.1 CRP Strategy: Model Structure Comparison

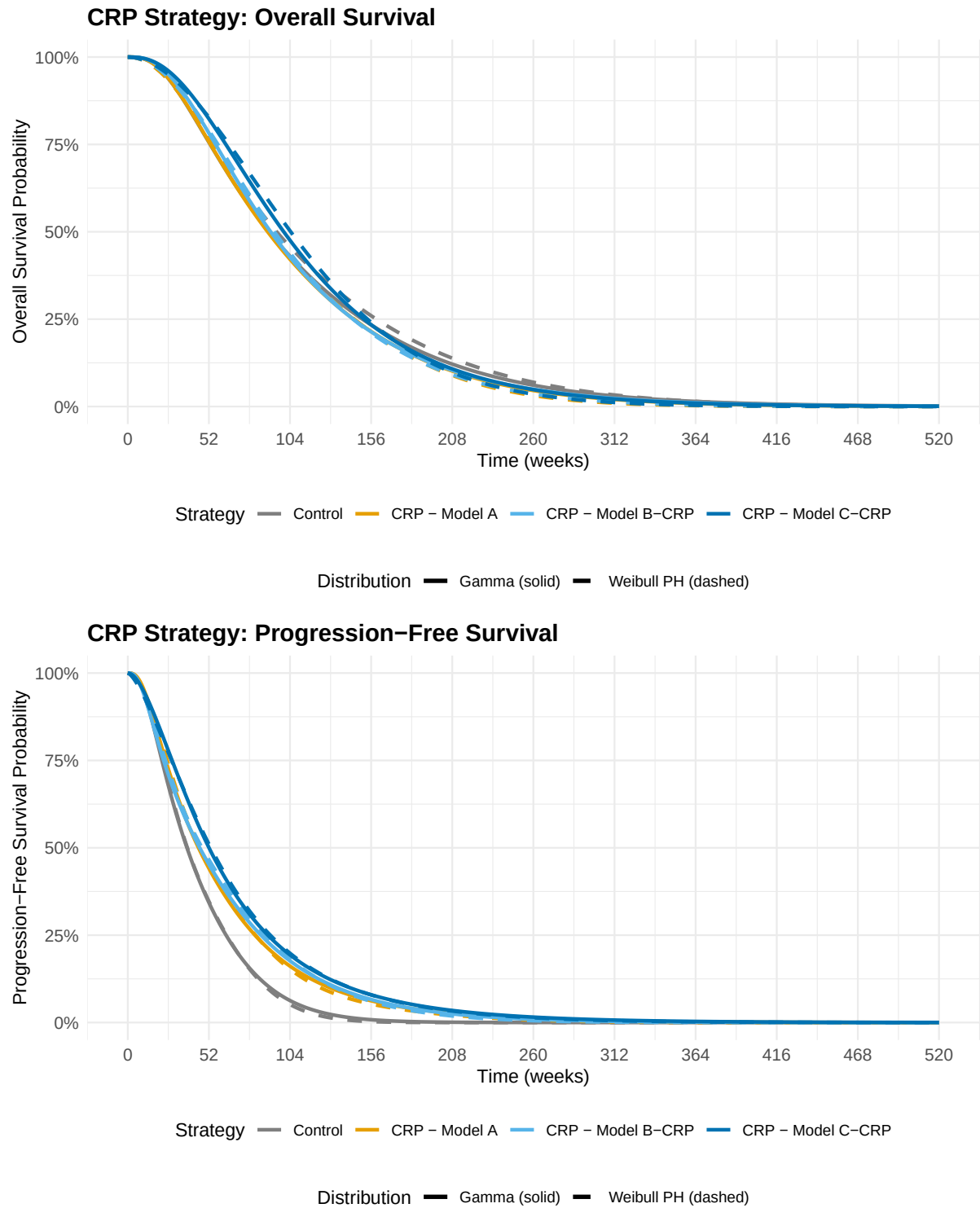


Figure 6: CRP strategy comparison across model structure types. Each line shows population-weighted survival for CRP-guided treatment allocation using different model specifications. Solid lines = Gamma, dashed lines = Weibull PH.

7.2 TLR Strategy: Model Structure Comparison

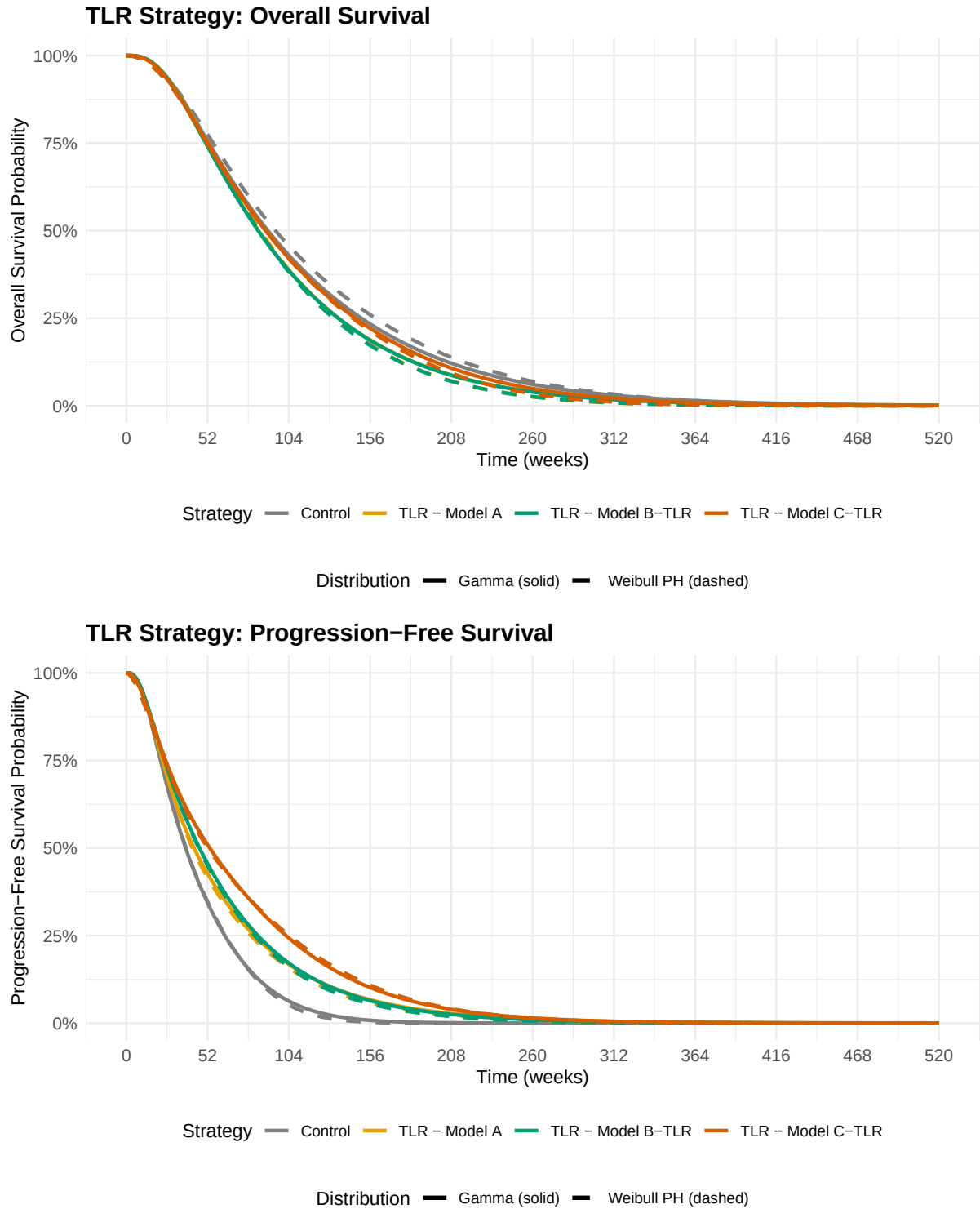


Figure 7: TLR strategy comparison across model structure types. Each line shows population-weighted survival for TLR-guided treatment allocation using different model specifications. Solid lines = Gamma, dashed lines = Weibull PH.

7.3 TMB_BRAF Strategy: Model Structure Comparison

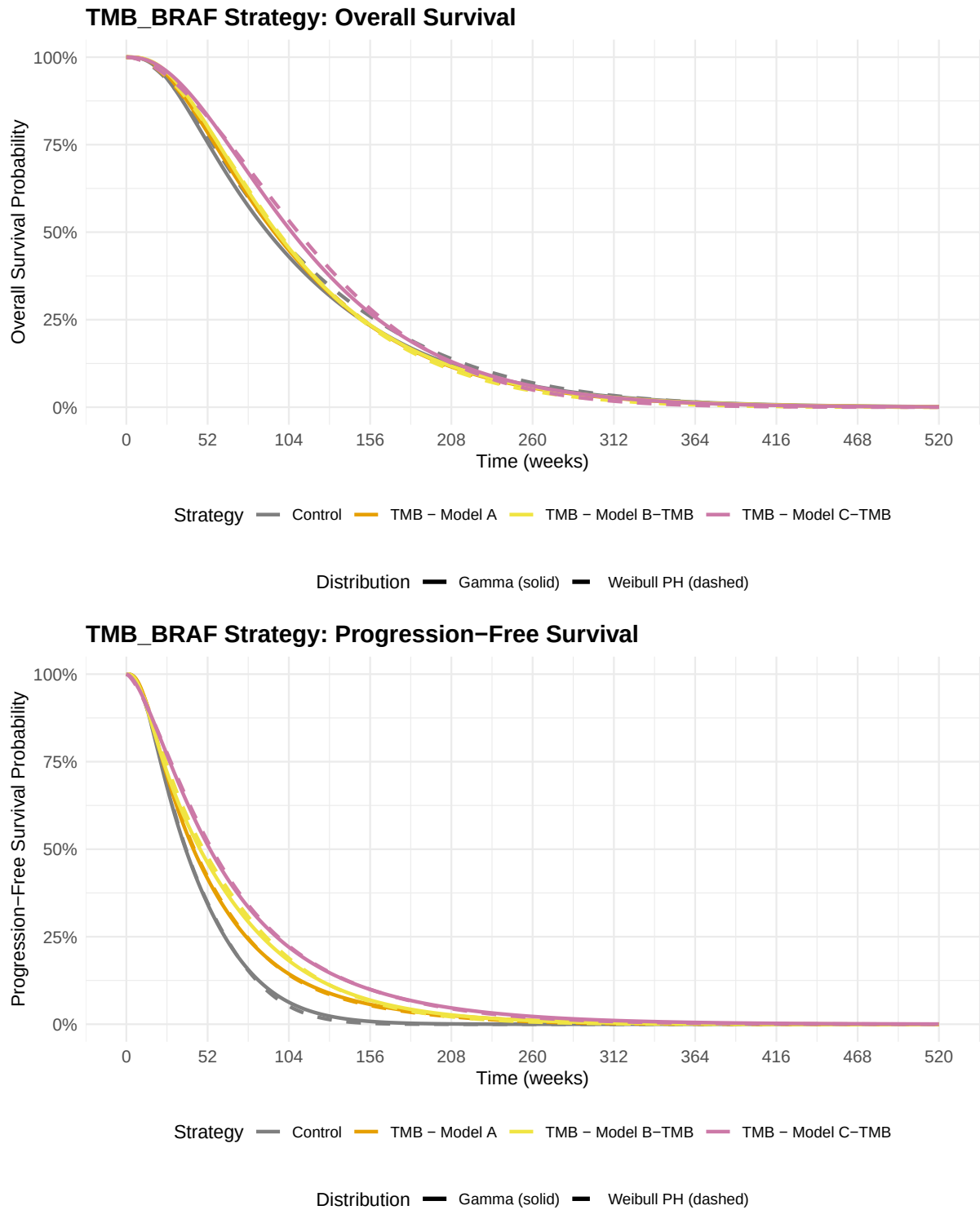


Figure 8: TMB_BRAF strategy comparison across model structure types. Each line shows population-weighted survival for TMB_BRAF-guided treatment allocation using different model specifications. Solid lines = Gamma, dashed lines = Weibull PH.

7.4 Life-Years by Strategy and Model Structure

The following table shows life-years (area under the survival curve) for each treatment strategy across all model structure types. This allows comparison of how model structure choice affects life-year estimates.

Table 32: Life-Years by Strategy and Model Structure Type (Gamma / Weibull PH)

Strategy	Model Type	OS LY (G/W)	PFS LY (G/W)	OS Gain	PFS Gain
Control	—	2.16 / 2.26	0.89 / 0.88	—	—
CRP	Model A	2.08 / 2.04	1.19 / 1.17	-0.08 / -0.21	+0.30 / +0.29
CRP	Model B-CRP	2.12 / 2.09	1.2 / 1.21	-0.04 / -0.17	+0.31 / +0.33
CRP	Model C-CRP	2.23 / 2.22	1.32 / 1.32	+0.07 / -0.04	+0.43 / +0.44
TLR	Model A	1.98 / 1.92	1.18 / 1.14	-0.18 / -0.33	+0.29 / +0.26
TLR	Model B-TLR	1.98 / 1.91	1.21 / 1.17	-0.19 / -0.34	+0.32 / +0.29
TLR	Model C-TLR	2.09 / 2.03	1.39 / 1.39	-0.08 / -0.23	+0.50 / +0.51
TMB_BRAF	Model A	2.19 / 2.17	1.14 / 1.13	+0.02 / -0.08	+0.24 / +0.25
TMB_BRAF	Model B-TMB	2.22 / 2.17	1.23 / 1.24	+0.06 / -0.08	+0.33 / +0.36
TMB_BRAF	Model C-TMB	2.35 / 2.34	1.4 / 1.4	+0.19 / +0.09	+0.51 / +0.52

Note: Life-years are calculated as the area under the survival curve (restricted mean survival time) over the model time horizon. G = Gamma distribution, W = Weibull PH distribution. Gains are relative to the Control strategy. Population-weighted values account for biomarker prevalence in each strategy.

8 Summary

This report compared three model structure types for parametric survival analysis:

- **Model A (Joint):** All biomarkers and interactions in one model - provides mutually adjusted estimates and is used for the primary cost-effectiveness analysis
- **Model B (Focused):** Each biomarker modeled with its treatment interaction while controlling for other biomarker main effects. CRP and TMB/BRAF share a formula with both interaction terms; TLR has a separate formula with only its interaction term. Allows biomarker-specific model selection
- **Model C (Single-biomarker):** Each biomarker modeled independently - simplest specification but may overestimate effects if biomarkers are correlated

8.1 Key Findings

- **Distribution selection:** Gamma distribution selected by AIC for both OS and PFS; Weibull PH retained as reference
- **Model fit:** All seven model structure types show reasonable fit based on AIC; Model A has lowest AIC (best fit) due to additional parameters
- **Interaction effects:** Treatment-biomarker interaction hazard ratios show [consistency/variation] across model types, indicating [robust/sensitive] effect estimates
- **Model structure impact:** Comparison across Models A, B, and C assesses sensitivity to biomarker correlation and model complexity
- **Validation:** Visual comparison with Kaplan-Meier estimates confirms adequate model fit across all parametric distributions
- **Clinical implications:** Model A (Joint) provides the most comprehensive assessment for cost-effectiveness analysis by accounting for all biomarker interactions simultaneously

8.2 Recommendations

- Use **Model A (Joint)** for primary cost-effectiveness analysis as it provides mutually adjusted biomarker interaction estimates
- Consider **Model B (Focused)** variants if biomarker-specific model selection is desired
- **Model C (Single-biomarker)** results should be interpreted cautiously if biomarkers are correlated, as effect estimates may be biased

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